# Analysis and Forecasting of Chilean Consumer Price Index using Time Series Models

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*Time Series Analysis and Forecasting*
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***Abstract*—This work studies the monthly Consumer Price Index, All items (PCPI_IX) series for Chile. The series was extracted from the file CPITimeSeries.csv, cleaned, and transformed into a monthly time series format. The analysis uses data from January 1990 to August 2022, with 392 observations and no internal missing values. First, a descriptive analysis of the series is carried out, including trend, seasonality, and stationarity. Then, three forecasting approaches are compared: a linear extrapolation baseline, a SARIMA model, and a triple exponential smoothing model. A rolling one-step-ahead evaluation is also performed, and a multivariate comparison with Mexico, Colombia, and Brazil is included using monthly inflation computed from the CPI. The results show that the TES model obtains the lowest error in the direct test forecast, although both main models underestimate the strong final increase. In contrast, the rolling approach greatly reduces the errors because it updates the available information before each prediction.**

***Index Terms*—time series, CPI, Chile, SARIMA, exponential smoothing, forecasting, multivariate analysis**

## I. Introduction

The objective of the project is to analyze and forecast a monthly CPI time series for a selected country. In this work, Chile was selected and the indicator *Consumer Price Index, All items*, with code PCPI_IX, was used. This variable does not represent individual prices. It is an aggregated index that summarizes the evolution of the general level of consumer prices. For this reason, the interest is not only in the value of the index in a specific month, but also in its behavior over time.

The Chilean series is suitable for applying the methods studied in the course because it is long, monthly, and has no internal gaps in the selected period. It also shows a clear increasing trend, some seasonal structure, and a final test period where the index increases more strongly. This makes it possible to evaluate whether the models can capture both the historical structure and more recent changes.

The work is organized progressively. First, the dataset is prepared and the temporal train, validation, and test split is defined. Then, trend, seasonality, and stationarity are studied. After that, forecasting models are fitted and compared. Finally, a multivariate comparison with other countries is performed using monthly inflation rates, since CPI levels are not directly comparable between countries when their scales are different.

## II. Dataset and Preprocessing

The original file is in wide format. Each row corresponds to a combination of country, indicator, and attribute, while the monthly columns contain the values of the series. To build the working series, Chile, the indicator PCPI_IX, and the attribute *Value* were filtered. Then, the monthly columns were transformed into two columns: date and CPI.

Table I
Summary of the Chilean CPI series from 1990.

| Feature | Value |
|---|---|
| Country | Chile |
| Indicator | PCPI_IX |
| Frequency | Monthly |
| Period used | 1990-01 to 2022-08 |
| Observations | 392 |
| Internal missing values | 0 |
| Minimum | 20.9256 |
| Mean | 69.3128 |
| Maximum | 125.6700 |

Table II
Temporal split used in the experiment.

| Set | Start | End | N |
|---|---|---|---|
| Train | 1990-01 | 2018-08 | 344 |
| Validation | 2018-09 | 2020-08 | 24 |
| Test | 2020-09 | 2022-08 | 24 |

Before modeling, it was checked that the real Chilean series starts in January 1970 and ends in August 2022. However, following the project requirement, the final analysis was restricted to the period starting in January 1990. In this final period, 392 monthly observations were obtained. No internal missing values, missing months, or duplicated dates were detected. The basic summary of the series from 1990 is shown in Table I.

The split was chronological. A total of 344 observations were used for training, 24 for validation, and 24 for testing. This decision avoids mixing future information with past information and leaves two complete years for validation and two complete years for final evaluation. Fig. 1 shows this split.

## III. Descriptive Analysis and Transformations

The original series presents a very clear increasing trend. Therefore, it does not seem reasonable to treat it directly as stationary. The series does not fluctuate around a constant mean level; instead, the index level increases persistently during the full period.

Several alternatives were compared to study the trend: polynomial fitting, moving average, and LOWESS. The moving

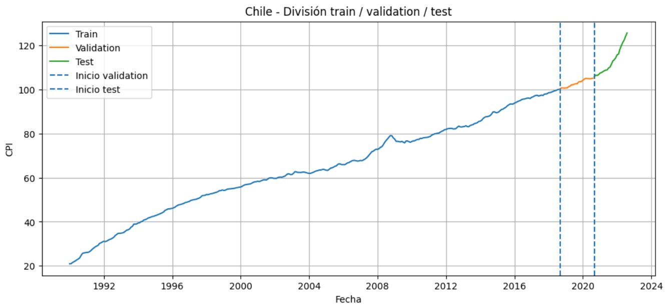


Figure 1. Chilean CPI series with train, validation, and test split.

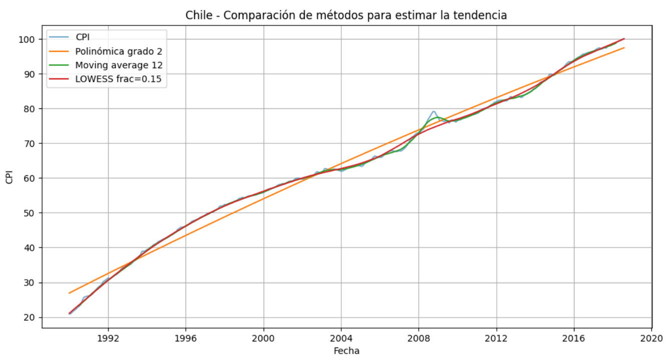


Figure 2. Trend estimation using moving average.

average was chosen as a simple descriptive option because it captures the general trend well and is easy to justify for a monthly series. Fig. 2 shows the trend estimation used in the exploratory analysis. The amplitude of the oscillations does not seem to increase proportionally with the CPI level, so an additive interpretation was considered reasonable as a first approach. Even so, this does not mean that the variance is perfectly constant, since there are periods with higher local variability.

After removing the trend, periodic patterns were analyzed. Although the autocorrelation suggested signals at lags close to six months, the frequency analysis indicated that the dominant frequency was one cycle per year. For that reason, an annual seasonal period was kept, with $S = 12$.

For the Box-Jenkins models, a regular first-order difference was applied to remove the trend, and a seasonal difference with $S = 12$ was also applied. The ADF test applied to the transformed series gave a statistic of $-5.5436$ and a p-value of $1.68 \cdot 10^{-6}$, so the null hypothesis of non-stationarity was rejected. Fig. 3 shows the differenced series and its ACF and PACF functions, which were used to guide the selection of SARIMA orders.

It is important to point out that a series can be stationary and still show structure in the ACF. This is not a contradiction. This temporal dependence is exactly what ARMA, ARIMA, or SARIMA models try to use.

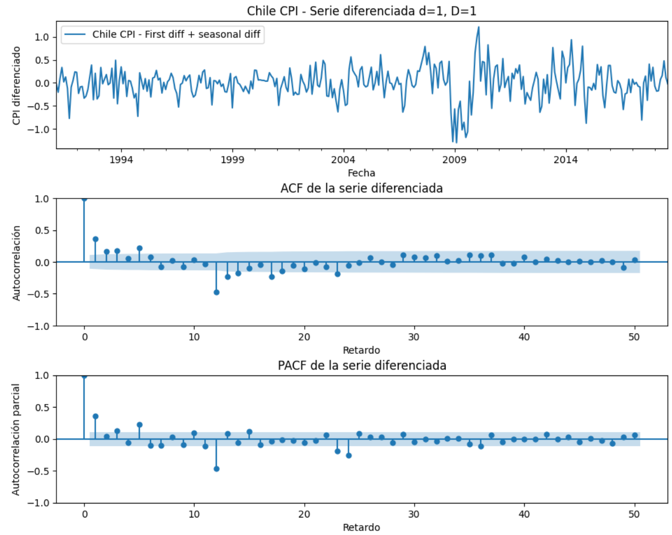


Figure 3. Differenced series and ACF/PACF analysis.

Table III
VALIDATION RESULTS OF THE SELECTED TES MODEL.

| Model | MSE | RMSE | MAE | MAPE |
|---|---|---|---|---|
| TES-m | 0.086 | 0.294 | 0.220 | 0.214% |

## IV. FORECASTING METHODOLOGY

Three main approaches were compared. The first one was a simple baseline based on trend extrapolation. This model does not try to capture the full temporal dynamics, but it provides a basic reference. Polynomial models of degree 1, 2, and 3 were tested using the training set and evaluated on validation. Degree 1 obtained the lowest RMSE in validation, so it was selected as the final baseline. Then, it was refitted using train and validation and evaluated on the test set.

The second approach was the Box-Jenkins methodology using SARIMA. The original series was not stationary, so $d = 1$, $D = 1$, and $S = 12$ were used. The final selected model was:

$$SARIMA(3, 1, 2) \times (1, 1, 1)_{12}.$$

This model was trained using train and validation to perform the final forecast on the test set.

The third approach was triple exponential smoothing (TES). The best model found in validation was:

$$TES\text{-}m(\alpha = 0.2, \gamma = 0.2, \delta = 0.3),$$

where the letter $m$ indicates a multiplicative formulation. In validation, this model obtained the results summarized in Table III. This validation step was used to choose the configuration before moving to the test set.

In addition to the direct 24-month forecast, a rolling one-step-ahead evaluation was applied. In this procedure, for each month of the test set, only the next value is predicted. Then, the real observed value is incorporated before moving to the

Table IV
BASELINE EVALUATION IN VALIDATION.

| Model | MAE | RMSE | MAPE |
|---|---|---|---|
| Polynomial degree 1 | 1.104 | 1.173 | 1.069% |
| Polynomial degree 2 | 3.283 | 3.332 | 3.185% |
| Polynomial degree 3 | 4.307 | 4.434 | 4.173% |

Table V
FINAL TEST RESULTS.

| Model | MSE | RMSE | MAE | MAPE | Cov. |
|---|---|---|---|---|---|
| Baseline | – | 7.587 | 5.974 | 5.091% | – |
| SARIMA | 46.796 | 6.841 | 5.046 | 4.270% | 54.167% |
| TES | 42.991 | 6.557 | 4.615 | 3.885% | 75.000% |

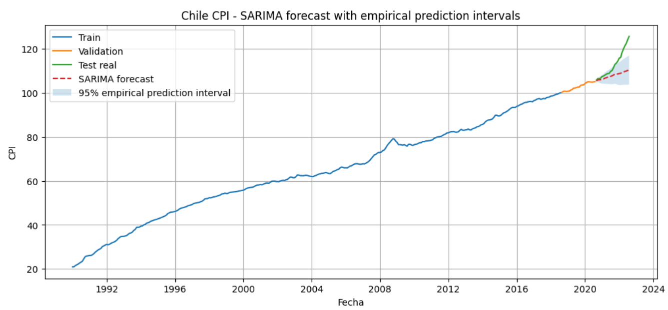


Figure 4. SARIMA multi-step forecast on the test set.

following month. This experiment does not represent the same problem as a direct 24-month forecast, but it shows how the models behave when they are updated with recent information.

## V. UNIVARIATE RESULTS

Table IV shows the validation results of the baseline. Although polynomial models of higher degree were tested, degree 1 was the best and also the simplest. This is important because a more flexible polynomial may fit some historical segments better but extrapolate worse into the future.

Table V summarizes the final test results for the three main models. The baseline obtains the worst result, as expected, because it only extrapolates a linear trend and does not use seasonal structure or temporal dependence. SARIMA improves the baseline, and TES obtains the lowest error among the direct forecasting models. However, the three models underestimate, to some extent, the strong growth observed at the end of the test period.

Fig. 4, Fig. 5, and Fig. 6 show the direct forecasts on the test set. Visually, the baseline follows a more rigid path. SARIMA and TES follow the general evolution better, but they do not fully capture the final acceleration of the CPI.

The rolling results are very different because the model is updated at each step with the previous real value. Table VI shows that rolling SARIMA reaches RMSE = 0.5369 and MAPE = 0.3656%, while rolling TES obtains RMSE = 1.4767 and MAPE = 1.0185%. In this scenario, SARIMA performs better than TES. This does not contradict the multi-step

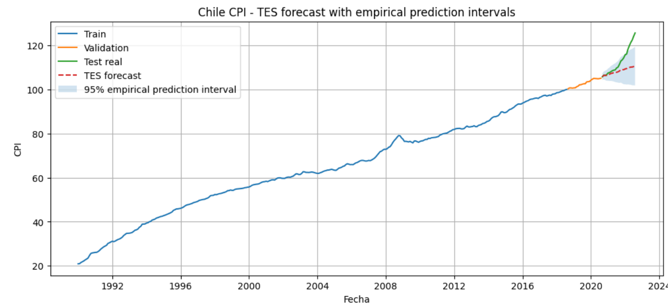


Figure 5. TES multi-step forecast on the test set.

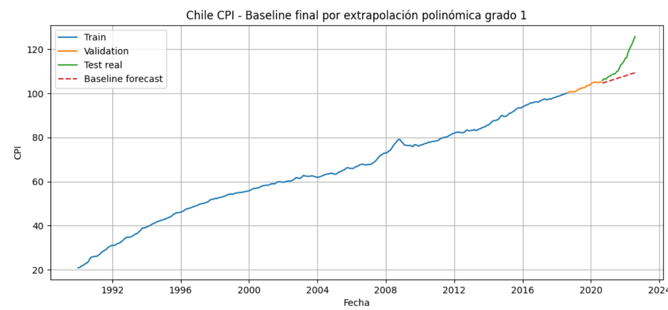


Figure 6. Polynomial baseline forecast on the test set.

Table VI
ROLLING ONE-STEP-AHEAD RESULTS ON THE TEST SET.

| Model | MSE | RMSE | MAE | MAPE |
|---|---|---|---|---|
| SARIMA rolling | 0.288 | 0.537 | 0.418 | 0.366% |
| TES rolling | 2.181 | 1.477 | 1.190 | 1.019% |

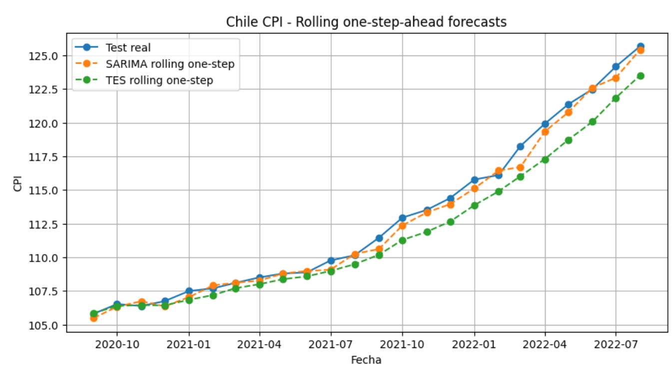


Figure 7. Rolling one-step-ahead comparison between SARIMA and TES.

comparison, because two different forecasting problems are being evaluated.

## VI. MULTIVARIATE ANALYSIS

The multivariate part was carried out by comparing Chile with Mexico, Colombia, and Brazil. First, it was observed that CPI levels were not directly comparable. Brazil, for example, had much higher values on the index scale. For this reason,

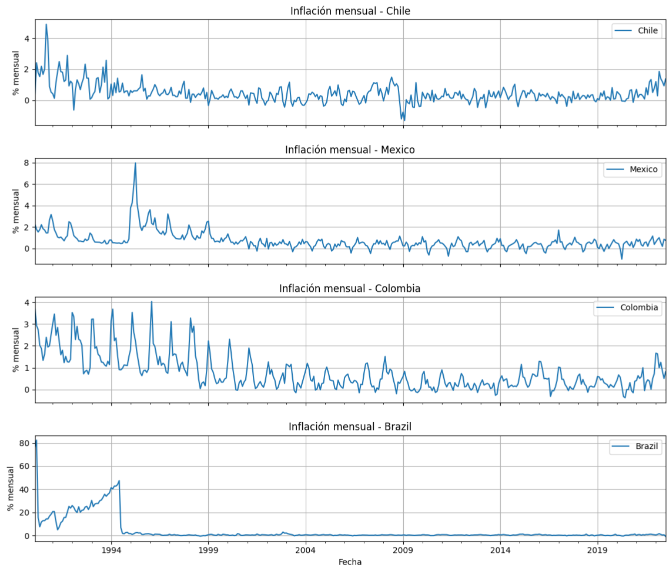


Figure 8. Monthly inflation computed from CPI for Chile, Mexico, Colombia, and Brazil.

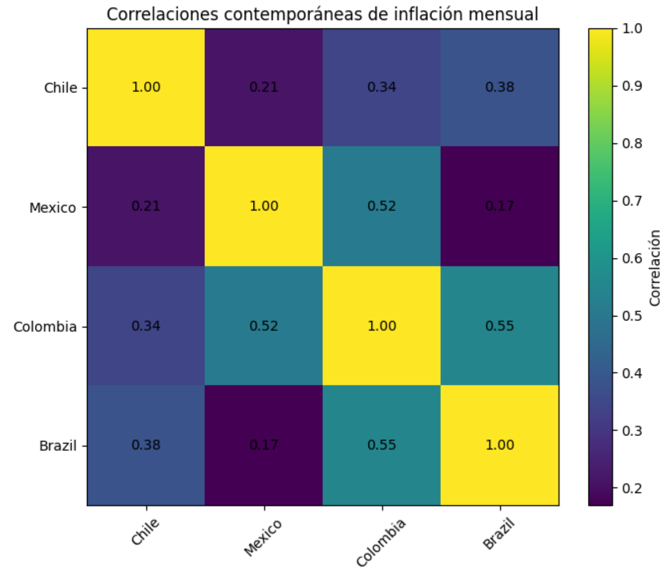


Figure 9. Contemporaneous correlation matrix of monthly inflation rates.

Table VII
SUMMARY OF MONTHLY INFLATION BY COUNTRY.

| Country | Mean | Std. | Min. | Max. |
|---|---|---|---|---|
| Chile | 0.459 | 0.614 | -1.302 | 4.892 |
| Mexico | 0.714 | 0.854 | -1.014 | 7.968 |
| Colombia | 0.776 | 0.825 | -0.376 | 4.029 |
| Brazil | 4.066 | 10.274 | -0.680 | 82.390 |

Table VIII
CROSS-CORRELATION SUMMARY WITH CHILE.

| Comparison | CCS lag 0 | Max. abs. | Lag |
|---|---|---|---|
| Chile–Mexico | 0.212 | 0.310 | 5 |
| Chile–Colombia | 0.343 | 0.542 | 6 |
| Chile–Brazil | 0.380 | 0.503 | -7 |

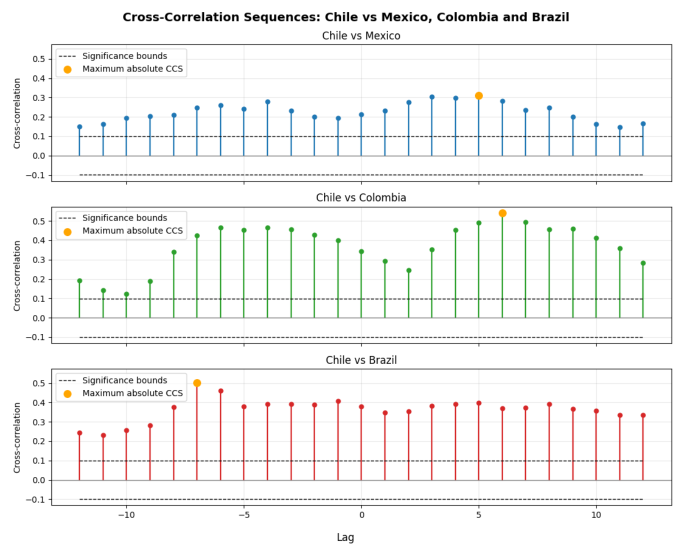


Figure 10. Cross-correlation between Chile and the compared countries.

instead of comparing levels, monthly inflation was computed as the percentage change of the CPI.

The common period used was from January 1990 to July 2022, with 391 observations and with a value lost in August in Brazil, although it was removed to avoid problems. Fig. 8 shows the monthly inflation rates of the four countries. Brazil stands out because it presents much higher variability, especially at the beginning of the period.

Table VII summarizes the basic statistics of monthly inflation. Chile has a monthly mean of 0.4595 and a standard deviation of 0.6140. Brazil has a much higher mean and standard deviation, which confirms that its dynamics are quite different during the analyzed period.

The contemporaneous correlation matrix was also computed. As shown in Fig. 9, the correlations between Chile and the other countries are positive, but not very high. Chile has a correlation of 0.213 with Mexico, 0.344 with Colombia, and 0.381 with Brazil. Therefore, Chilean inflation shares some relationship with the others, but it also keeps its own dynamics.

Finally, cross-correlations between Chile and the other countries were analyzed. This analysis allows us to study whether the relationship appears in the same month or with lags. Table VIII summarizes the main values. Chile-Colombia shows the highest absolute correlation, with 0.5416 at lag 6. Chile-Brazil reaches 0.5026 at lag -7. These results suggest temporal associations, but they should not be interpreted as causality.

## VII. DISCUSSION

The analysis confirms that the Chilean CPI series should not be treated directly as stationary. The increasing trend is strong and the original ACF decreases slowly. After regular and seasonal differencing, the ADF test indicates stationarity, which justifies the use of SARIMA with $d = 1$, $D = 1$, and $S = 12$. This decision is consistent with what was observed in the notebook: the regular difference reduces the trend and the seasonal difference helps to treat the annual pattern.

One important decision was to keep $S = 12$ even though some plots showed signals close to six months. The reason is that the series is monthly and the frequency analysis showed an annual cycle as the main component. Therefore, the semiannual peaks were interpreted carefully as part of the oscillatory structure, but not as enough evidence to change the main seasonal period to six months.

In the direct 24-month forecast, TES obtains the best overall results, followed by SARIMA and then the baseline. This shows that including level, trend, and seasonality clearly improves the results compared with a simple linear extrapolation. However, neither SARIMA nor TES fully captures the strong growth observed during the test period. The most careful interpretation is that the models learned previous historical patterns, but the final segment contains a stronger increase that was not fully represented in the data used to fit the models.

The baseline is useful even though it has the worst result. Its purpose is not to compete in complexity with SARIMA or TES, but to serve as a minimum reference. If an advanced model does not clearly improve the baseline, then its complexity is not justified. In this case, SARIMA and TES do improve the baseline in RMSE, MAE, and MAPE, which indicates that using temporal dependence and seasonality adds useful information.

The comparison between SARIMA and TES must also be read carefully. TES is better in the multi-step forecast because it generates a trajectory closer to the complete test set. SARIMA is slightly behind in that case. However, this conclusion changes when rolling one-step-ahead is used. The rolling experiment updates the model with each new real observation before predicting the next month. For this reason, it represents a more favorable scenario, but also a more realistic one if the forecast is updated month by month.

The rolling experiment changes the interpretation. By updating the model at each step with the previous real observation, the error decreases a lot. Rolling SARIMA obtains better metrics than rolling TES. This means that the SARIMA model adapts well when the forecast is very short-term and the information is continuously updated. Therefore, it cannot be said that one model is always better: TES was better for the direct 24-month horizon, while SARIMA was better in the rolling scenario.

The coverage of the intervals also gives information. In the final table, TES has a coverage of 75%, higher than SARIMA, which has 54.167%. This means that, in the test set, the TES bands contain more real observations than the SARIMA bands. Even so, neither coverage reaches 95%, so the intervals do not fully capture the uncertainty of the final period.

The multivariate part provides a complementary view. When monthly inflation rates are compared, Chile shows positive relationships with Mexico, Colombia, and Brazil, but they are not strong enough to say that they move in the same way. Cross-correlations make it possible to detect associations with lags, especially with Colombia and Brazil, although the analysis itself does not prove causality. This part should be understood as a comparison and exploration of relationships, not as a causal demonstration between countries.

Table IX
MAIN DECISIONS AND INTERPRETATION USED.

| Decision | Careful interpretation |
|---|---|
| Initial additive model | The amplitude does not seem to grow proportionally with the level, but perfectly constant variance is not claimed. |
| $S = 12$ | The dominant frequency is annual, although there are signals at other lags. |
| Differencing $d = 1, D = 1$ | It is used to obtain a series more suitable for SARIMA. |
| Rolling | It evaluates one-step prediction with updating, not the same problem as multi-step forecasting. |
| CCS | It suggests temporal association, but it does not prove causality. |

## VIII. LIMITATIONS AND ANALYSIS DECISIONS

The report was written from the results of the notebook. Therefore, the conclusions are limited to the tests, models, and metrics calculated there. No additional external variables are used to explain the Chilean CPI, and no multivariate forecasting model with exogenous variables is fitted. The multivariate part is used as a comparison between countries, not as a causal model.

One important limitation is that the univariate models only use the past of the same series. If a strong change in behavior appears during the test period, the models may remain below the real series. This was observed in the direct forecast, where SARIMA, TES, and the baseline underestimate the final increase. This limitation does not mean that the models are poorly fitted; it means that the 24-month horizon is demanding when recent dynamics change.

It is also important to distinguish between CPI levels and monthly inflation. The CPI in levels was the main variable of the project and was used for the forecasting models. In contrast, monthly inflation was used in the multivariate part because it allows a better comparison between countries with very different index scales. This distinction avoids an incorrect interpretation of CPI levels among Chile, Mexico, Colombia, and Brazil.

Table IX summarizes some relevant decisions of the work and the careful way of interpreting them. These decisions help avoid statements that are too strong, for example saying that the variance is perfectly constant or that cross-correlation implies causality.

Finally, the metrics should be interpreted together with the plots. A low error in a table does not always explain the behavior of the model by itself. In this project, the plots clearly show that the main problem of the multi-step forecast is the acceleration in the final segment, while the rolling metrics show the benefit of updating the model month by month.

## IX. Conclusion

This work analyzed and forecasted the monthly PCPI_IX series for Chile from 1990 to 2022. The preprocessing produced a clean monthly series with no internal gaps. The characterization showed an increasing trend, an annual seasonal component, and the need to transform the series before applying stationary models.

In the direct forecast on the test set, the best model was multiplicative TES with $\alpha = 0.2$, $\gamma = 0.2$, and $\delta = 0.3$, with MAPE = 3.885%. SARIMA also improved the baseline, but it was slightly behind TES in the multi-step evaluation. In contrast, in the rolling one-step-ahead evaluation, SARIMA obtained the best result, with MAPE = 0.3656%. This shows that the choice of the best model depends on the type of forecasting problem: direct multi-month prediction or month-by-month updated prediction.

The multivariate comparison showed that Chilean monthly inflation rates have positive but moderate relationships with other Latin American countries. This part helped to contextualize the Chilean dynamics, although without turning correlations into causal conclusions. The comparison also showed that Brazil has a much more volatile dynamic in the analyzed period, so its values should be interpreted with special care.

As a general conclusion, the project shows that good forecasting does not depend only on fitting a model. It also requires preparing the series correctly, separating train, validation, and test properly, checking stationarity, using a baseline, and analyzing the errors. As future work, the multivariate comparison could be extended with models that explicitly use external variables, always checking whether they really improve the forecast compared with the univariate models.